

Operator Discovery Report: det_restart_controller_two_cluster_aware_fast_growth

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart controller that triggers early on stagnation and ramps restarts moderately; enables perturbation-based restarts to escape two-cluster/bottleneck traps while capping total restart count for 2-opt sta
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:3:9:2.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 5f3ba17d1d262af9.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Does not transplant cleanly into clustered_local_search.
- Runtime inflation is high at 2.139159.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 254.750957, distance-eval delta 9233.142857.
- cheapest_insertion_2opt gap delta -0.003457, runtime delta 1126.381271, distance-eval delta 8004.571429.
- random_restart_2opt gap delta 0.0, runtime delta -149.1348, distance-eval delta -7750.285714.
- sparse_three_opt gap delta 0.033767, runtime delta 61.559129, distance-eval delta 4034.142857.
- clustered_local_search gap delta 0.006887, runtime delta -37.965157, distance-eval delta -1834.0.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.083393 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.002615.
- Runtime inflation: 2.139159.
- Distance-evaluation inflation: 1.934645.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.002615.
- fixed_restart_budget mean gap 0.118702 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.118702 and delta vs full operator 0.0.